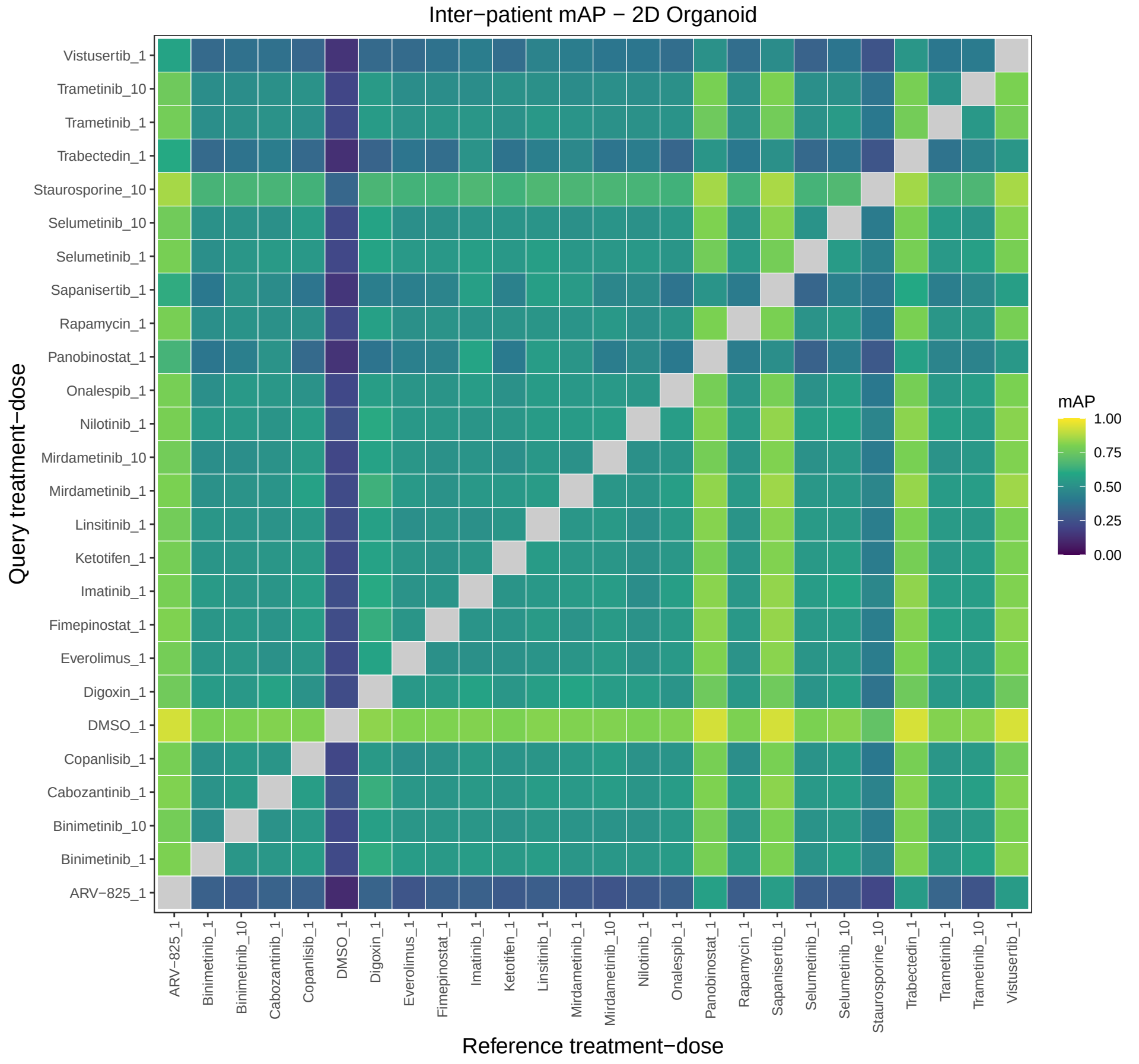
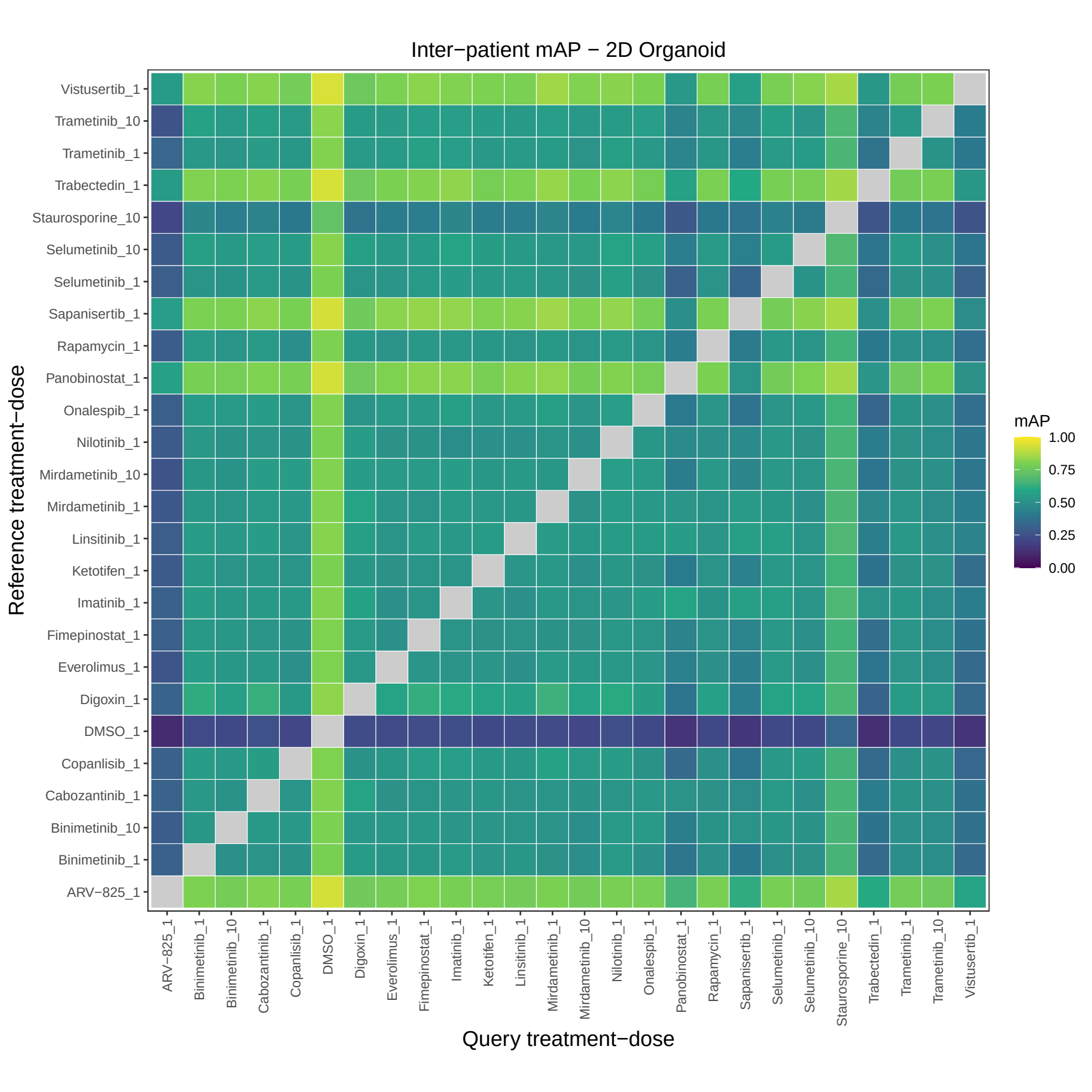
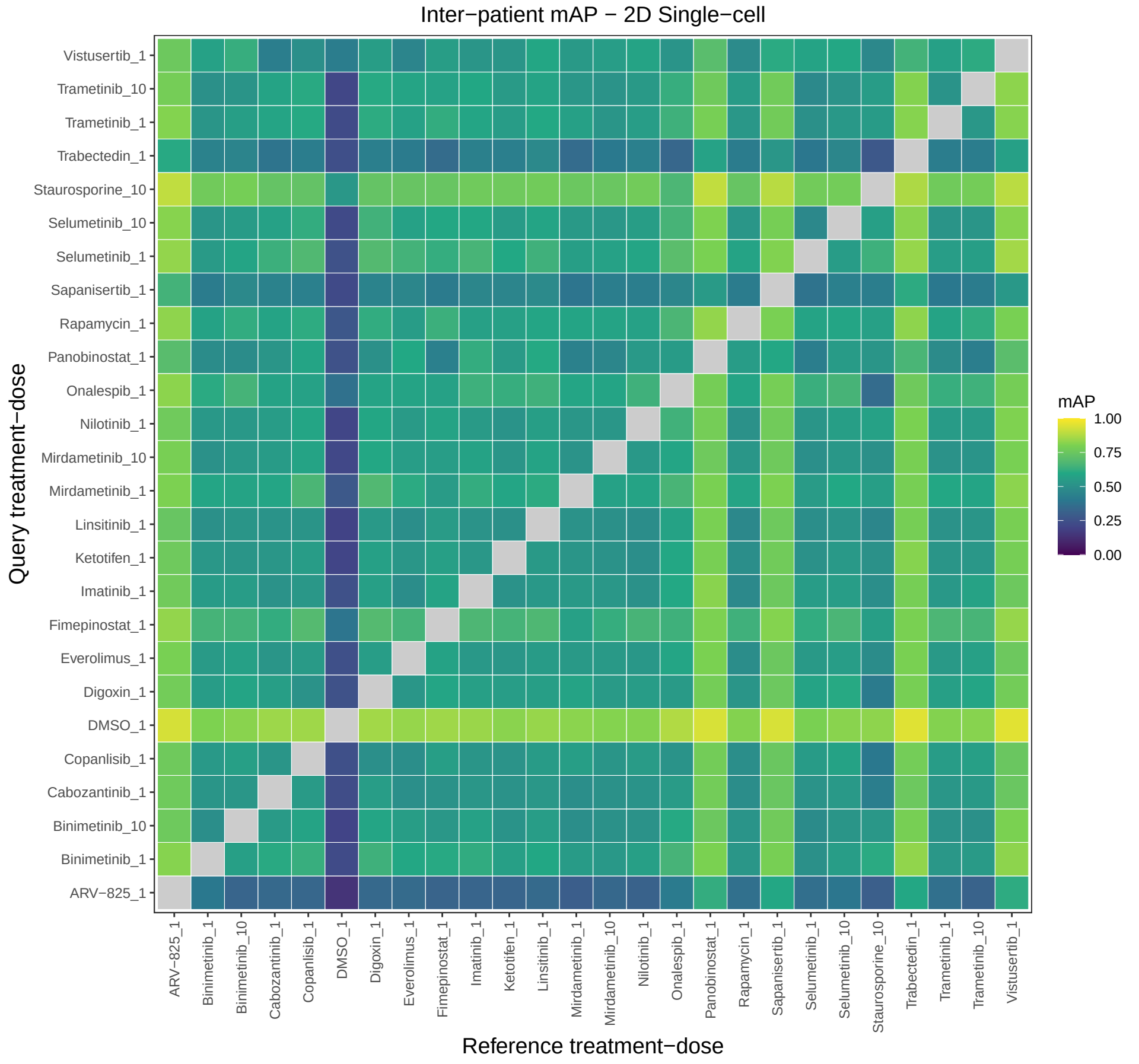
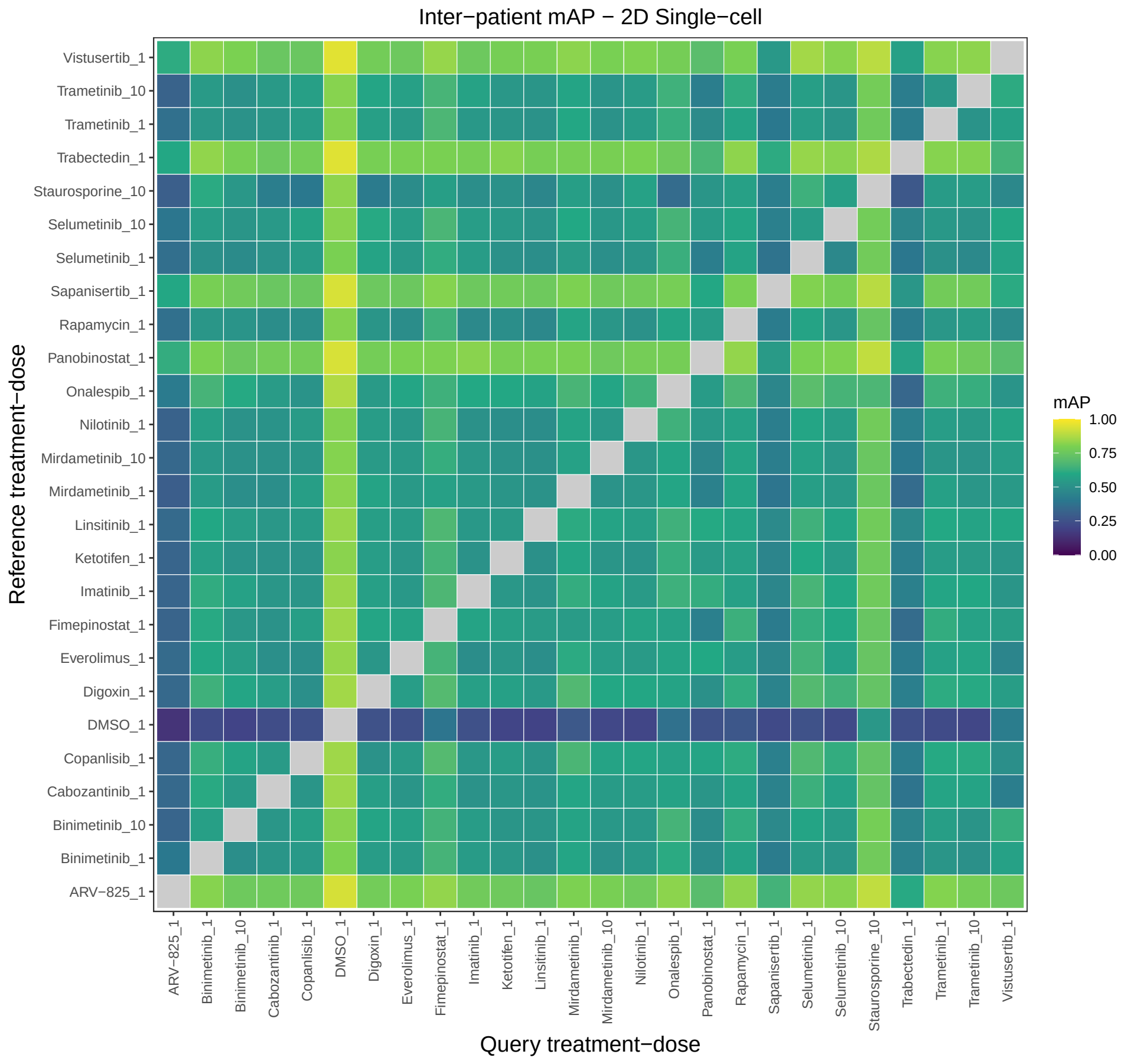
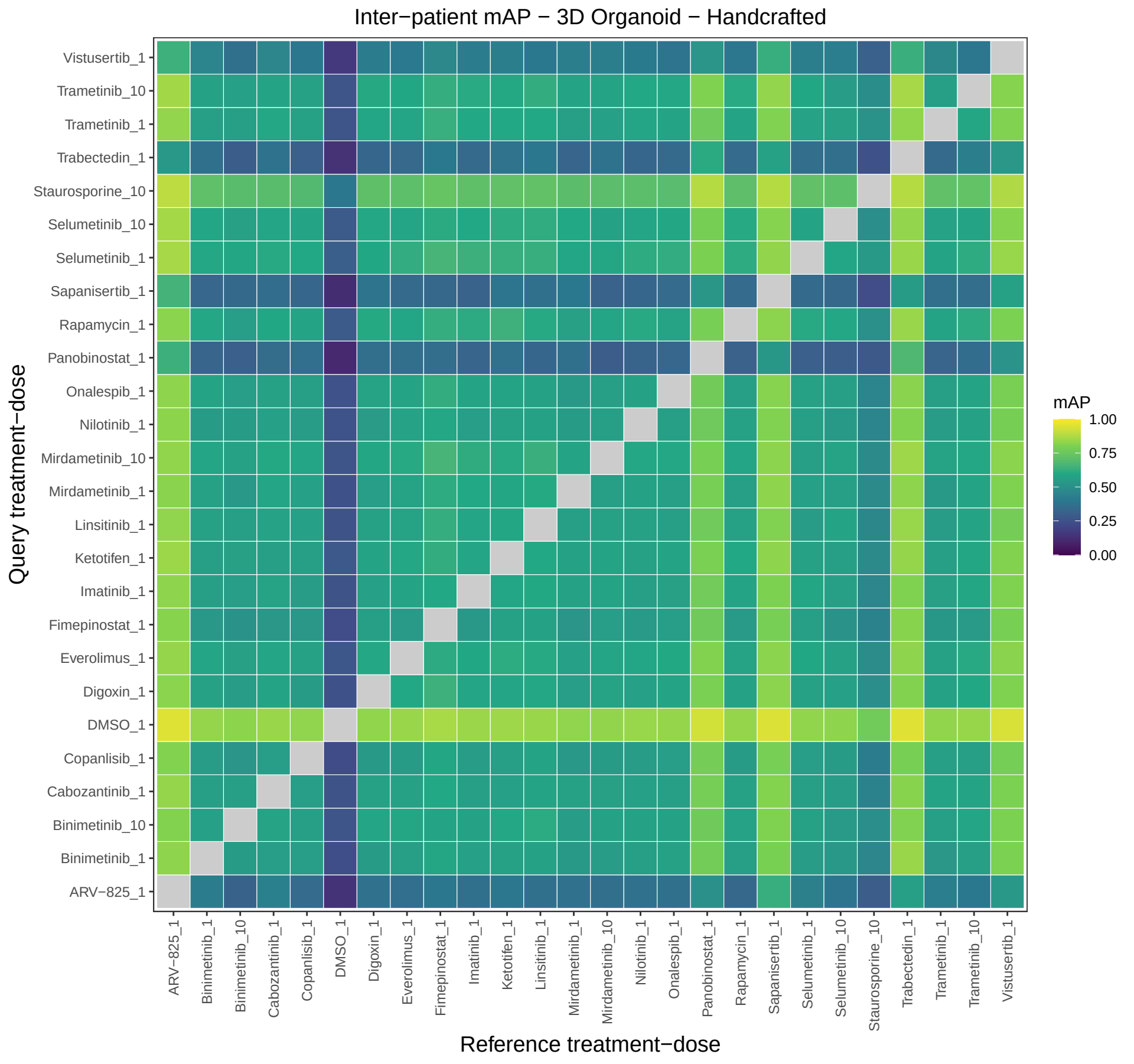
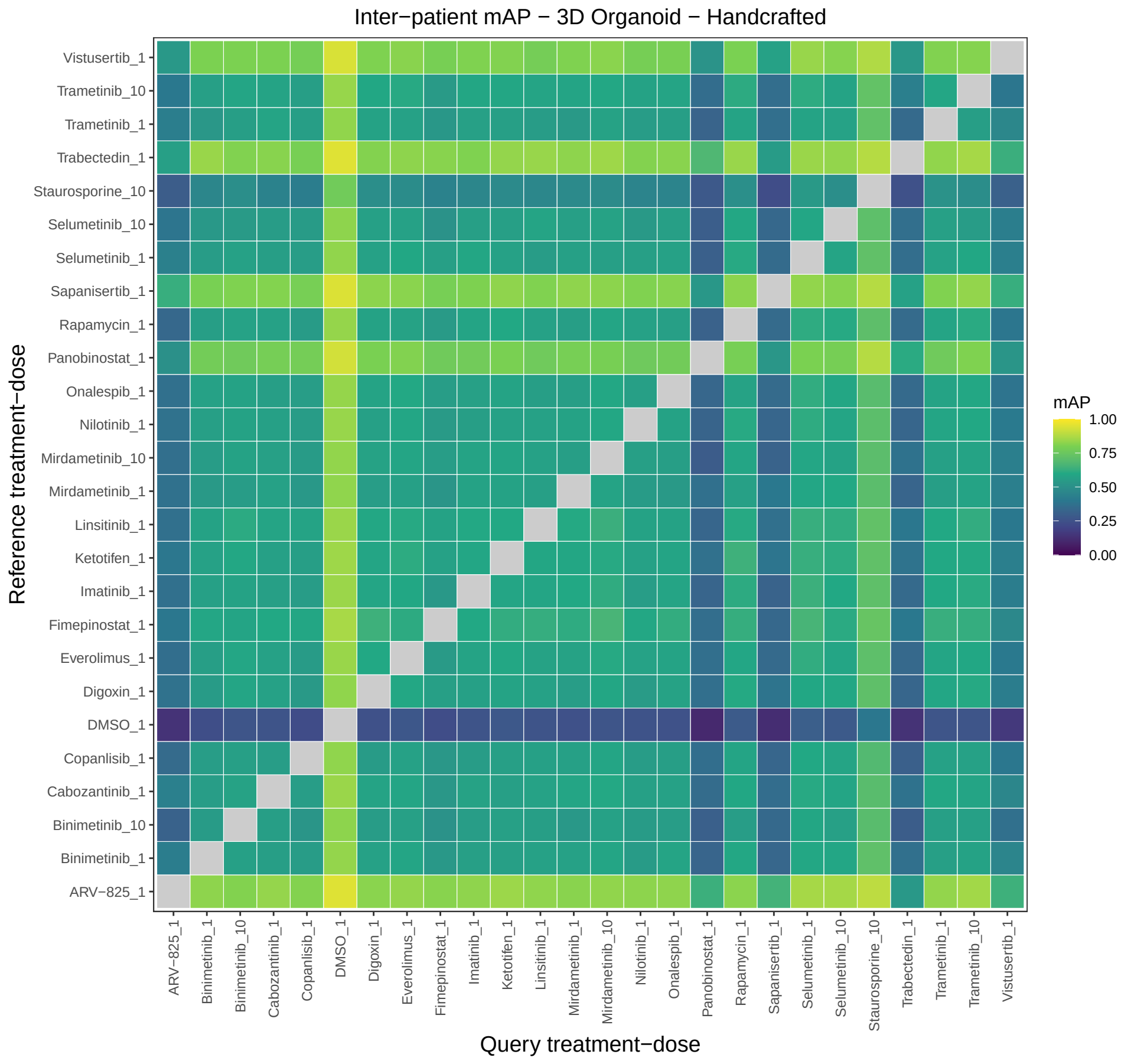

Inter−patient mAP − 2D Organoid
Query treatment−dose
Reference treatment−dose
mAP
1.00
0.75
0.50
0.25
0.00
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1

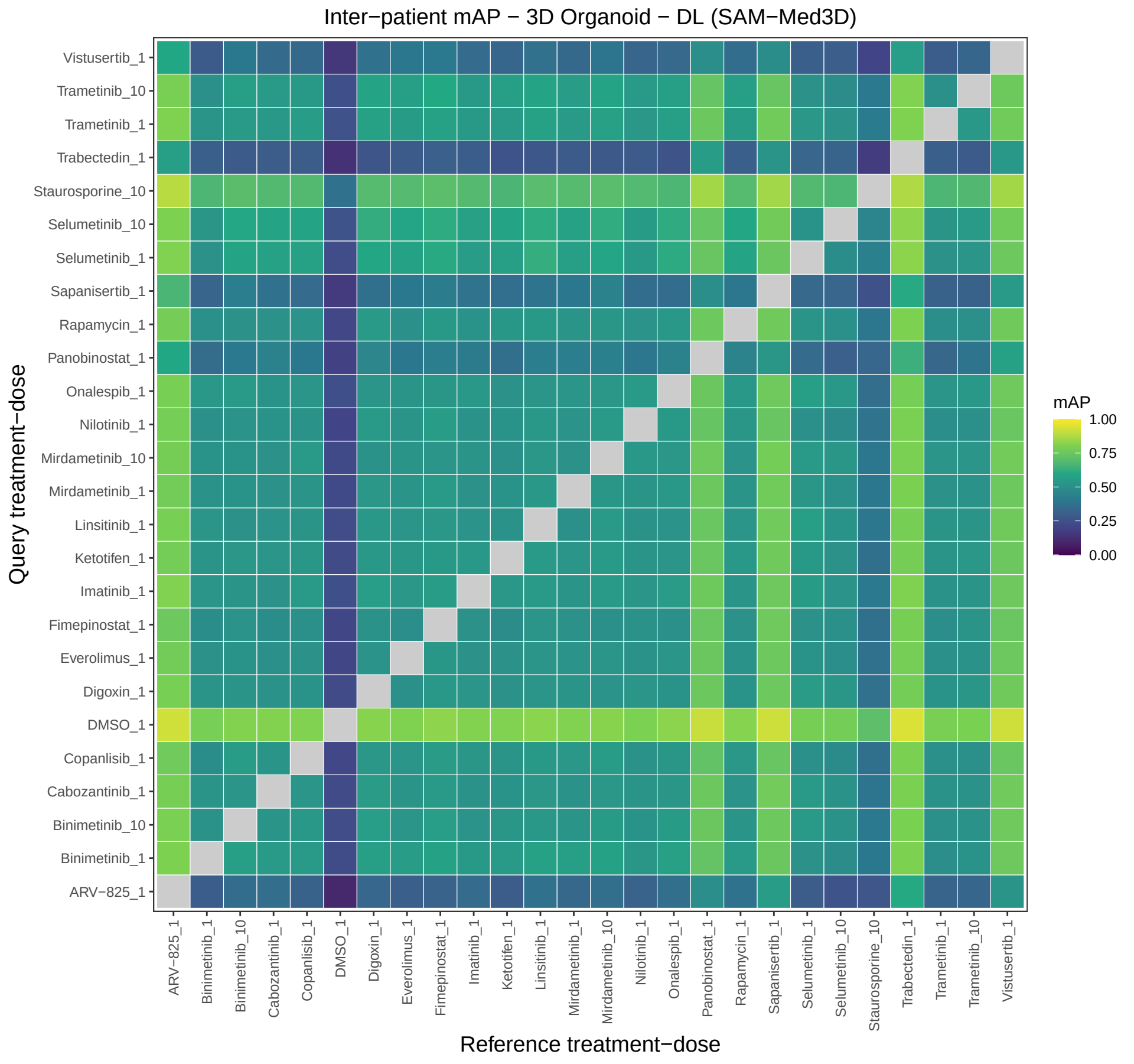

Inter−patient mAP − 2D Organoid
Reference treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Query treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00

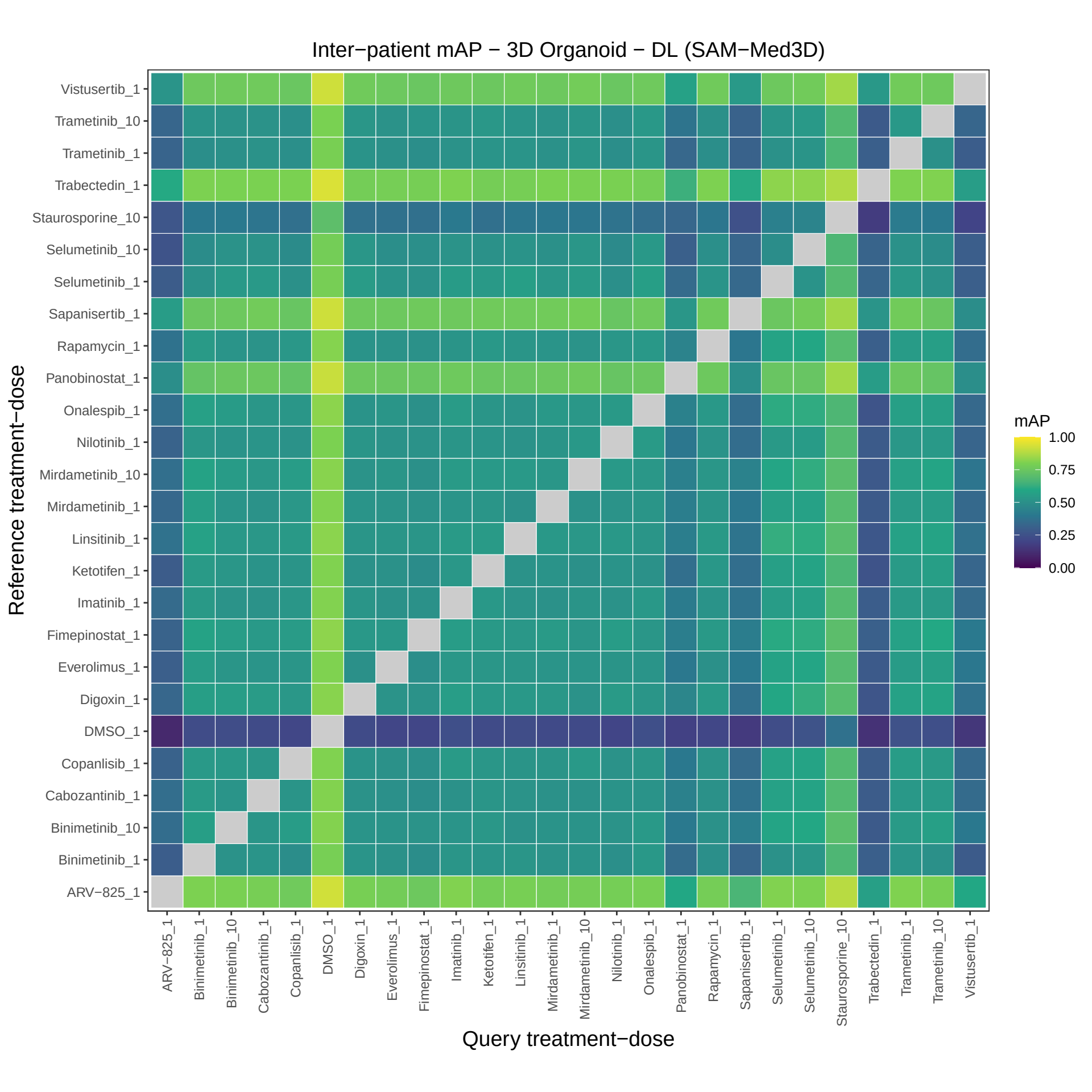

Inter−patient mAP − 2D Single−cell
Query treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Reference treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00

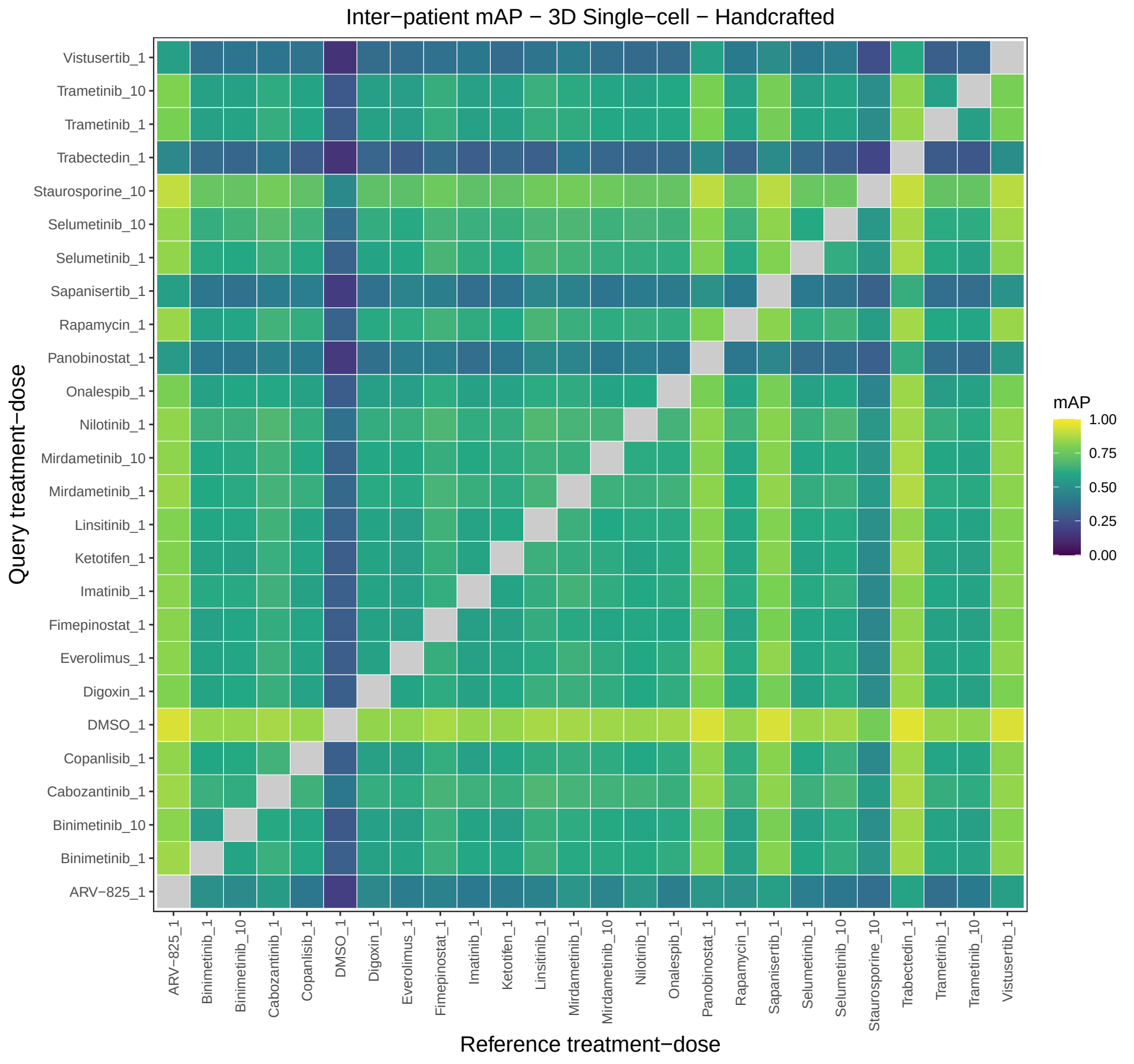

Inter−patient mAP − 2D Single−cell
Reference treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
Query treatment−dose
mAP
1.00
0.75
0.50
0.25
0.00

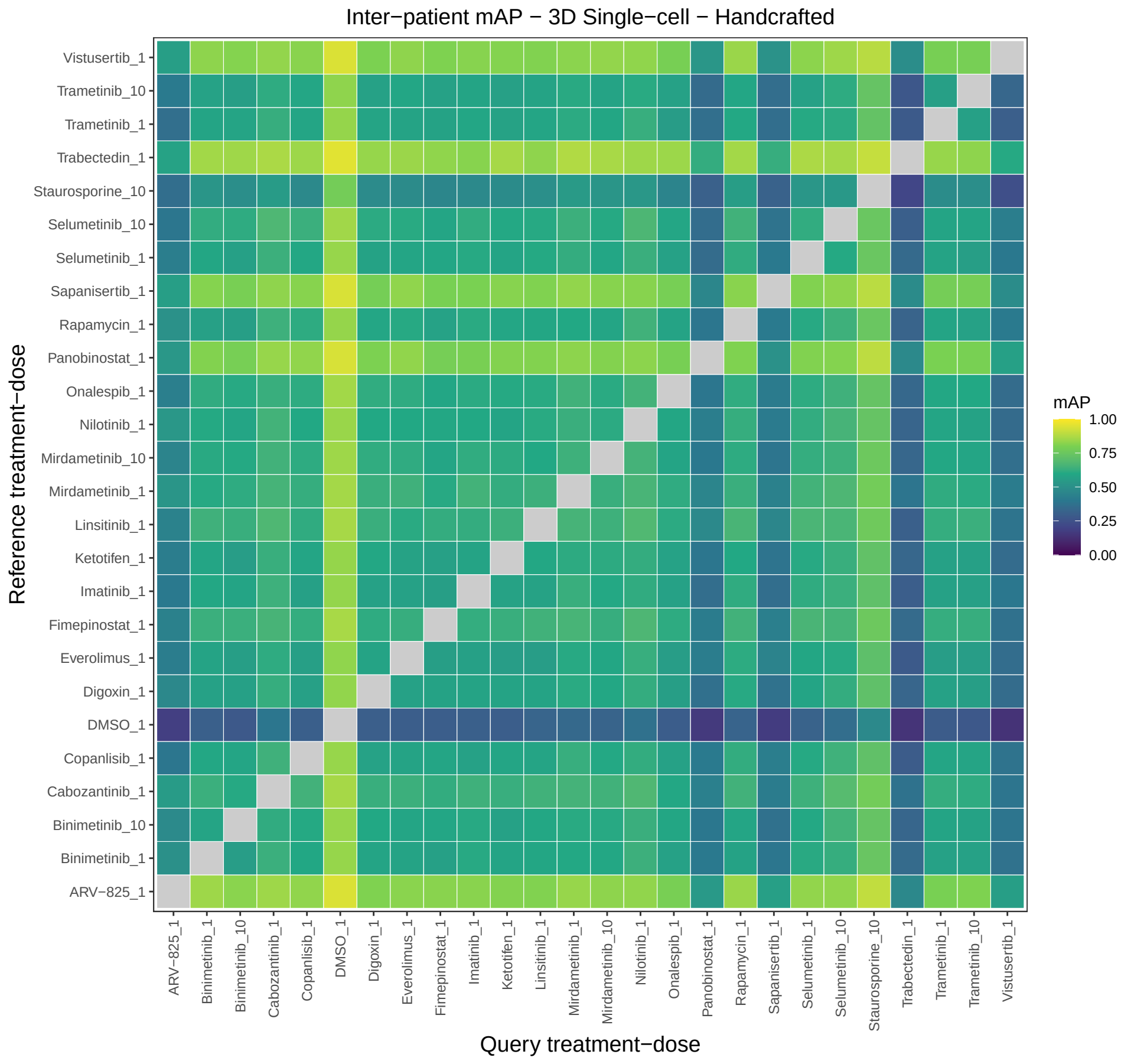

Inter−patient mAP − 3D Organoid − Handcrafted
Query treatment−dose
Reference treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
mAP
1.00
0.75
0.50
0.25
0.00

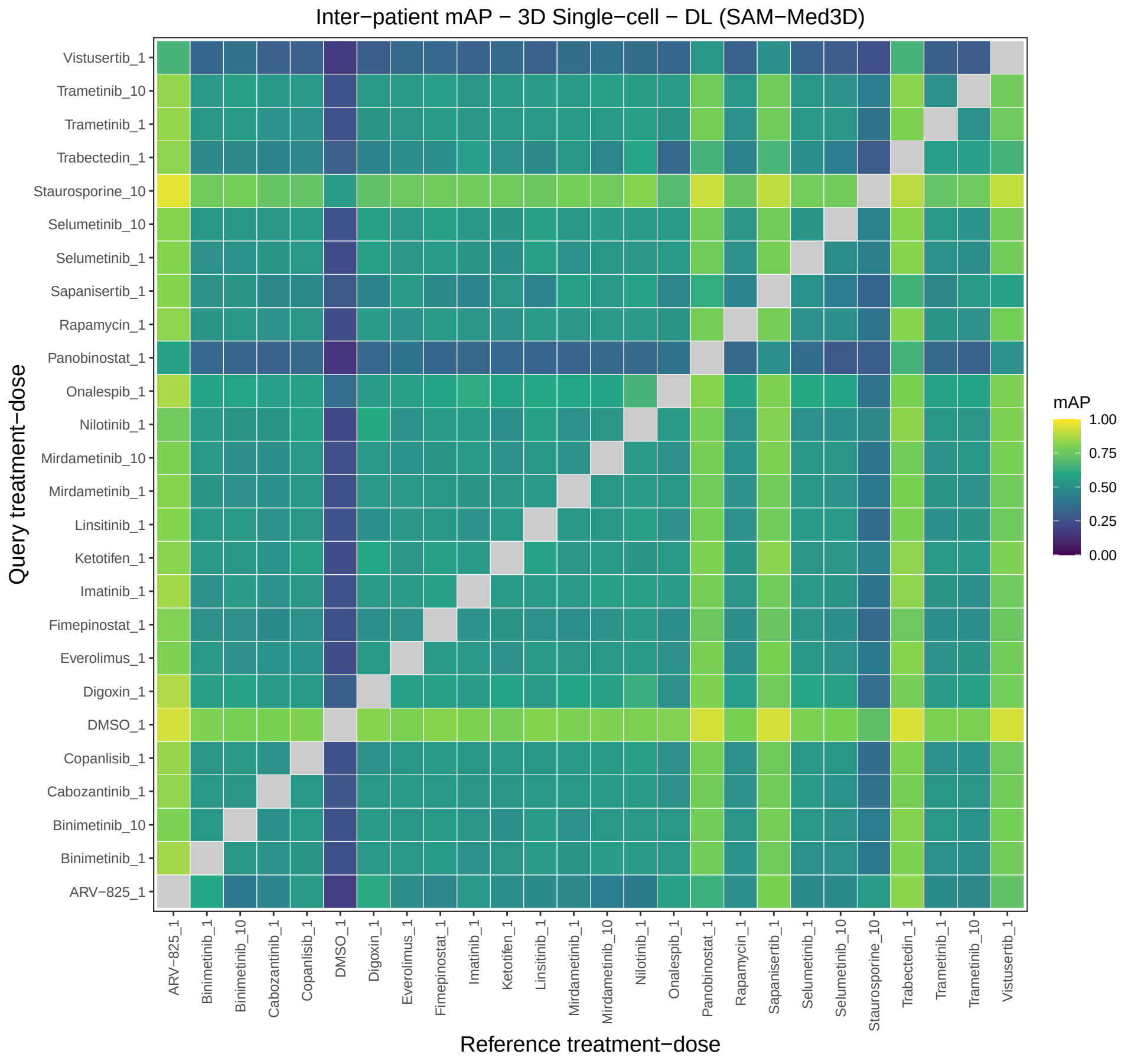

Inter−patient mAP − 3D Organoid − Handcrafted
Reference treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
Query treatment−dose
mAP
1.00
0.75
0.50
0.25
0.00

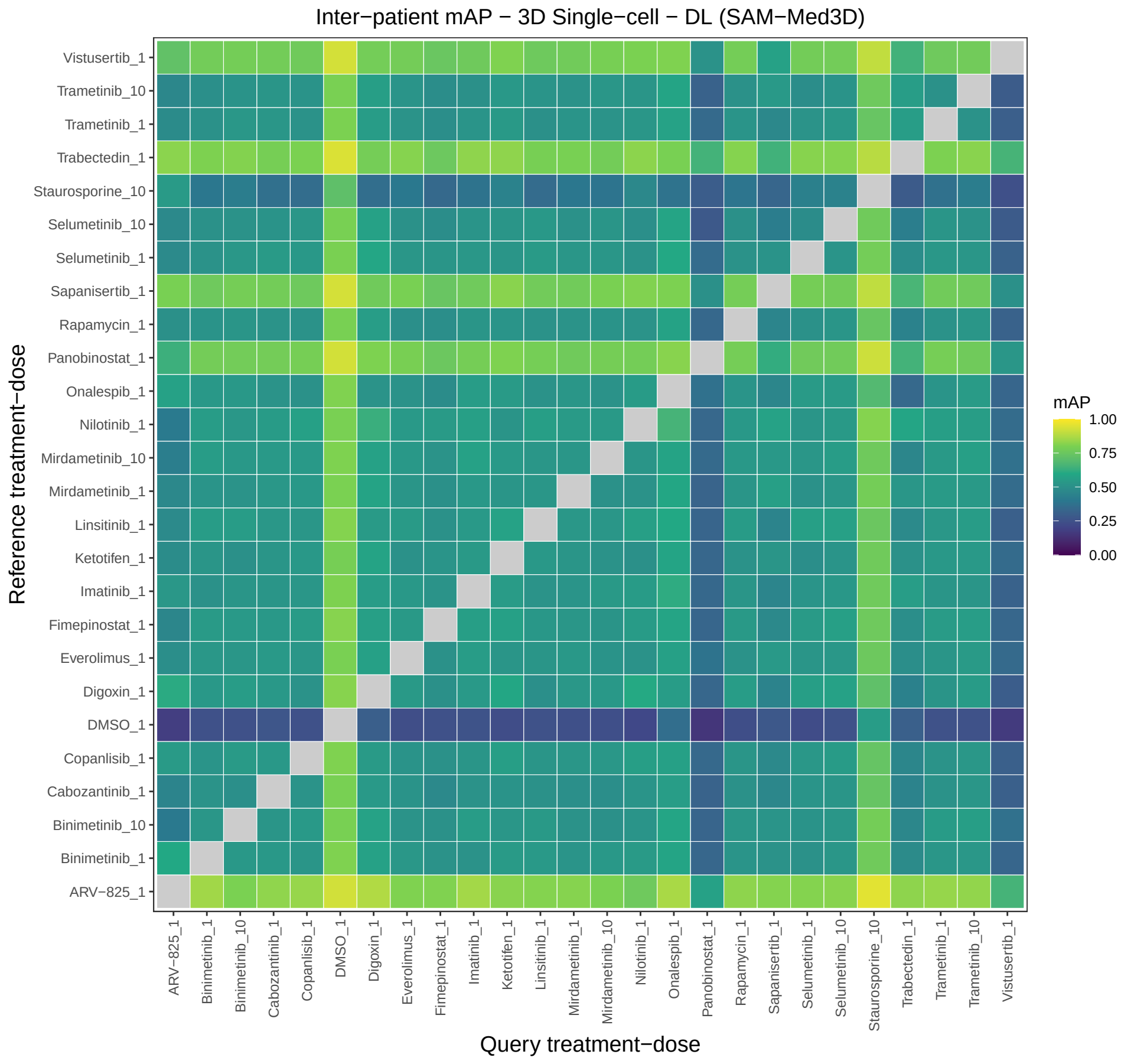

Inter−patient mAP − 3D Organoid − DL (SAM−Med3D)
Query treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Reference treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00

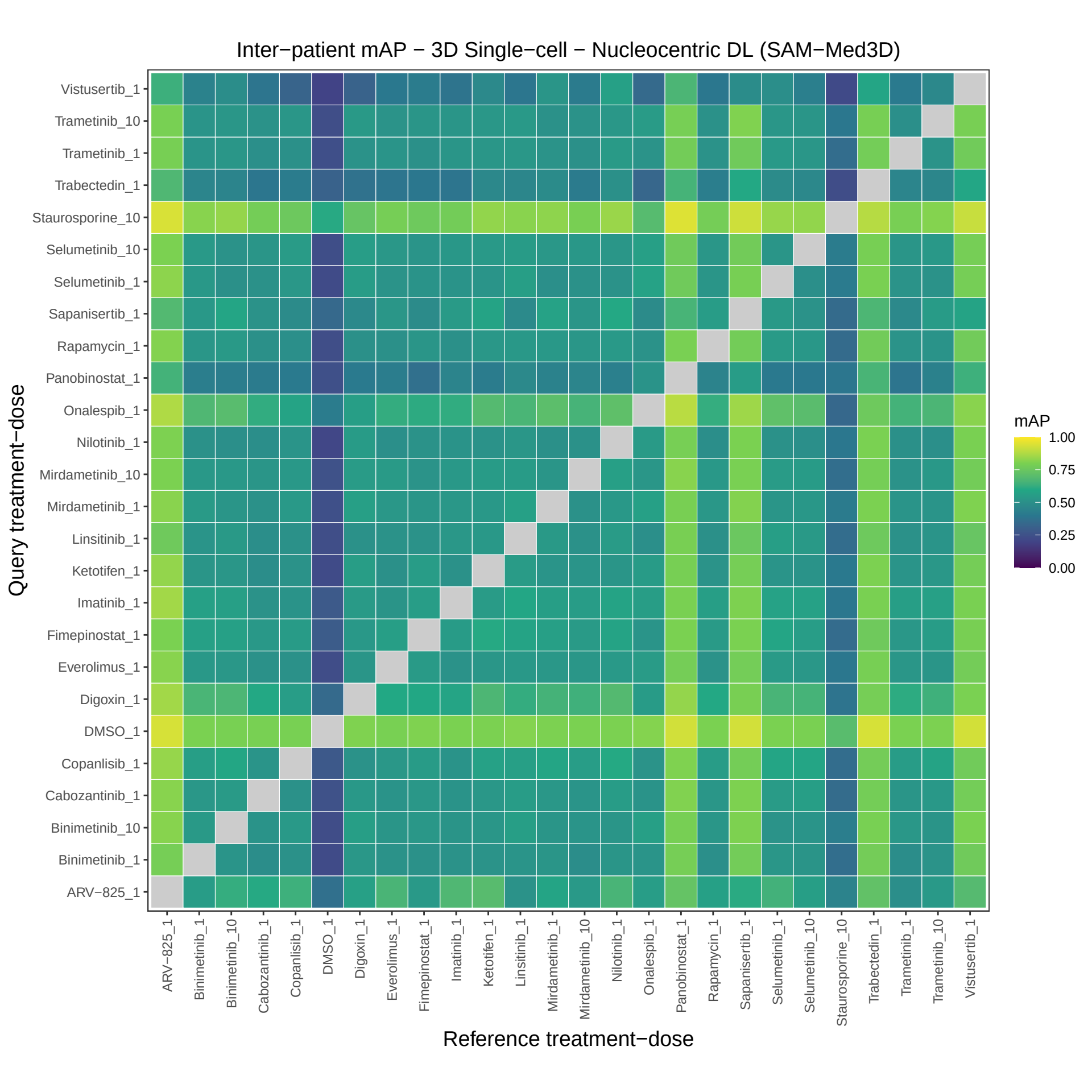

Inter−patient mAP − 3D Organoid − DL (SAM−Med3D)
Reference treatment−dose
Query treatment−dose
mAP
1.00
0.75
0.50
0.25
0.00
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1

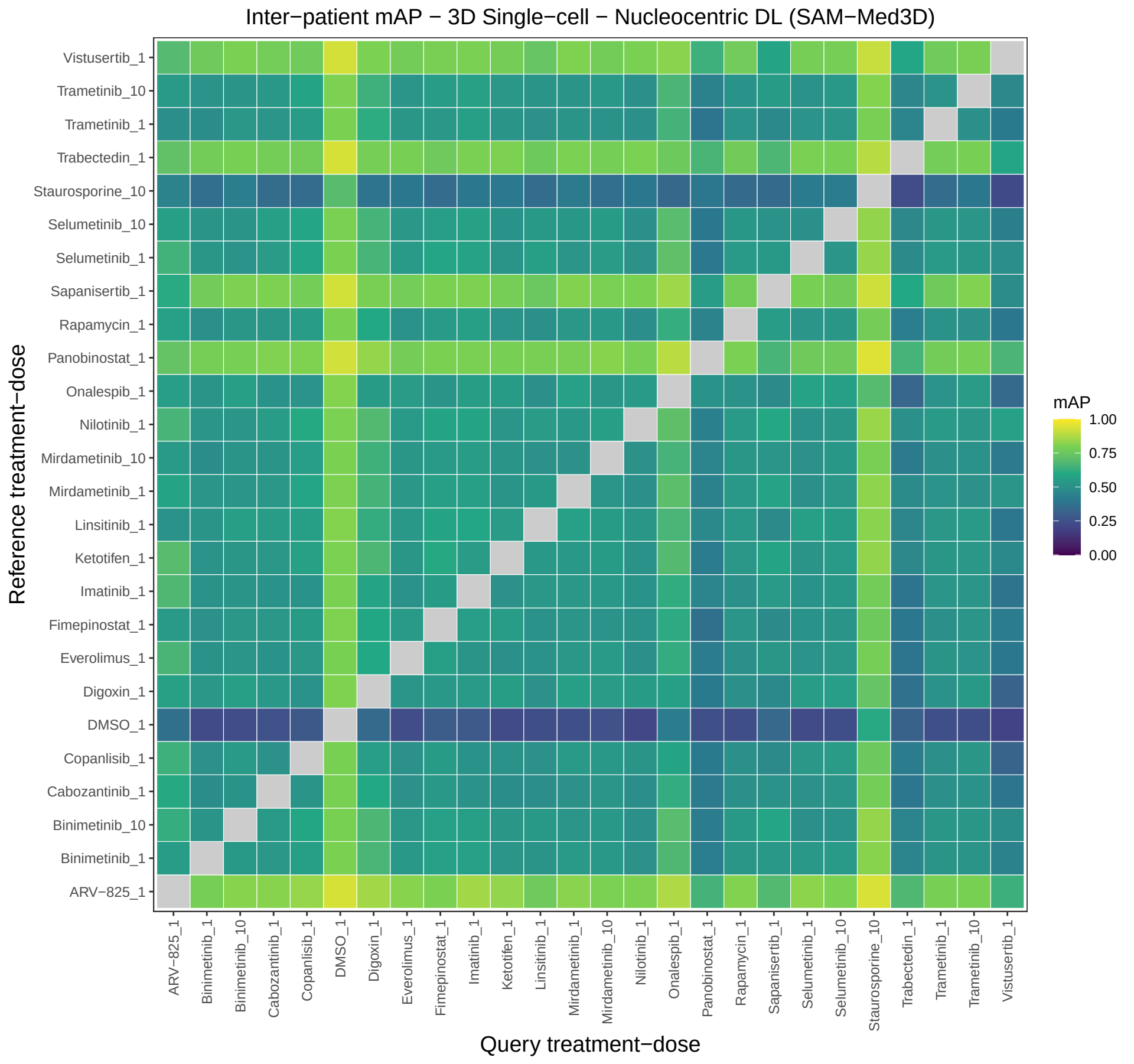

Inter−patient mAP − 3D Single−cell − Handcrafted
Query treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
Reference treatment−dose
mAP
1.00
0.75
0.50
0.25
0.00

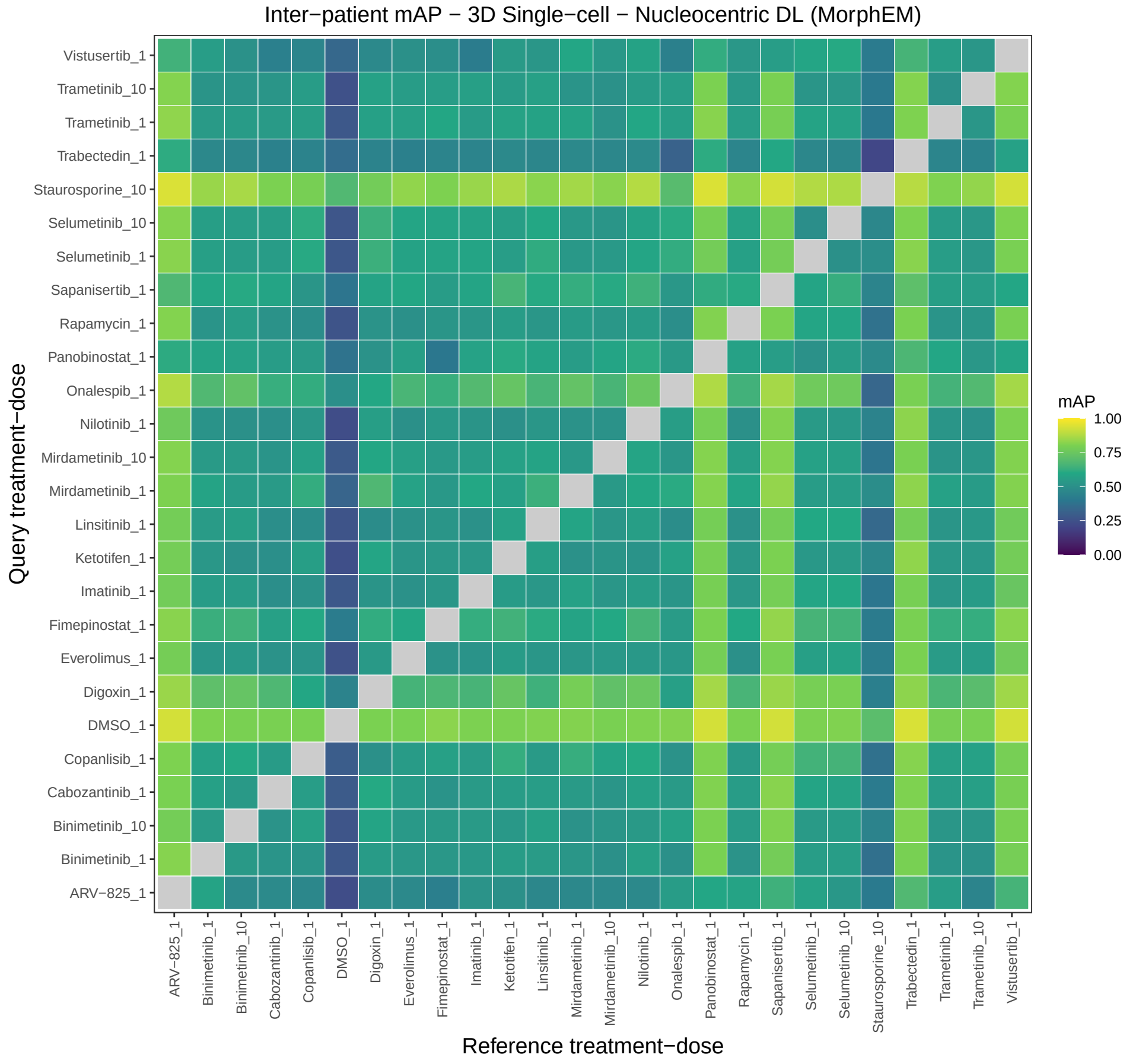

Inter−patient mAP − 3D Single−cell − Handcrafted
Reference treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Query treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00

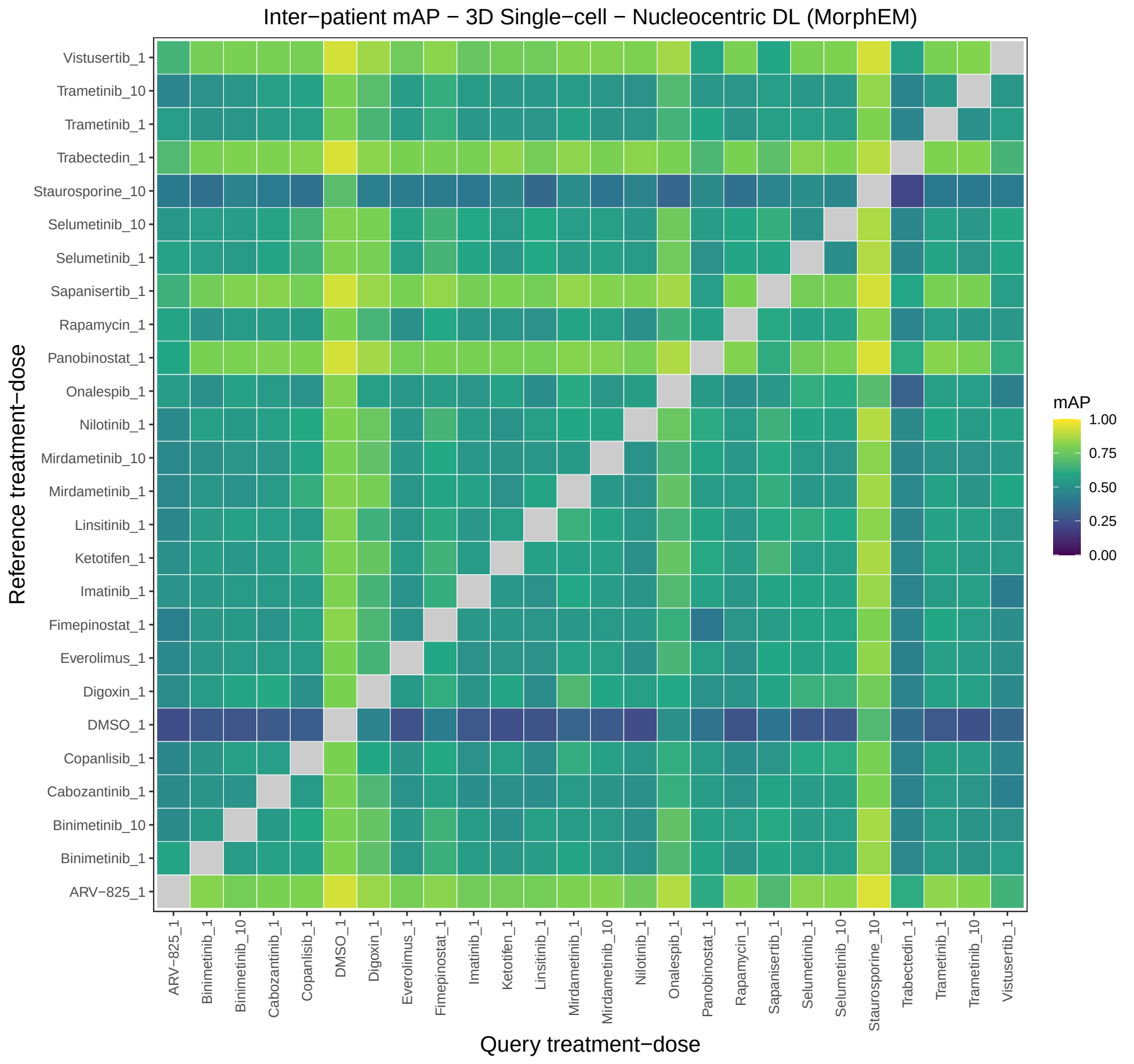

Inter−patient mAP − 3D Single−cell − DL (SAM−Med3D)
Query treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Reference treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00


Inter−patient mAP − 3D Single−cell − DL (SAM−Med3D)
Reference treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Query treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00


Inter−patient mAP − 3D Single−cell − Nucleocentric DL (SAM−Med3D)
Query treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Reference treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00


Inter−patient mAP − 3D Single−cell − Nucleocentric DL (SAM−Med3D)
Reference treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Query treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00


Inter−patient mAP − 3D Single−cell − Nucleocentric DL (MorphEM)
Query treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Reference treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00


Inter−patient mAP − 3D Single−cell − Nucleocentric DL (MorphEM)
Reference treatment−dose
Vistusertib_1
Trametinib_10
Trametinib_1
Trabectedin_1
Staurosporine_10
Selumetinib_10
Selumetinib_1
Sapanisertib_1
Rapamycin_1
Panobinostat_1
Onalespib_1
Nilotinib_1
Mirdametinib_10
Mirdametinib_1
Linsitinib_1
Ketotifen_1
Imatinib_1
Fimepinostat_1
Everolimus_1
Digoxin_1
DMSO_1
Copanlisib_1
Cabozantinib_1
Binimetinib_10
Binimetinib_1
ARV−825_1
Query treatment−dose
ARV−825_1
Binimetinib_1
Binimetinib_10
Cabozantinib_1
Copanlisib_1
DMSO_1
Digoxin_1
Everolimus_1
Fimepinostat_1
Imatinib_1
Ketotifen_1
Linsitinib_1
Mirdametinib_1
Mirdametinib_10
Nilotinib_1
Onalespib_1
Panobinostat_1
Rapamycin_1
Sapanisertib_1
Selumetinib_1
Selumetinib_10
Staurosporine_10
Trabectedin_1
Trametinib_1
Trametinib_10
Vistusertib_1
mAP
1.00
0.75
0.50
0.25
0.00